

D639

Technology for Instruction and Online Pedagogy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

An online middle school science teacher, whose course uses a new learning platform, posts weekly videos but never joins discussions or gives feedback beyond grades. Students say they feel alone. The course has 90 students. Which presence is weakest?

- A. The platform is the problem.
- B. Teaching presence, with social presence also suffering.
- C. Cognitive presence only.
- D. None; posting videos is full teaching presence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Students in an online course post icebreaker introductions and reply warmly to each other. Which presence does this build most directly?

- A. Assessment presence.
- B. Cognitive presence.
- C. Social presence.
- D. Teaching presence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A high school teacher of an online course wants students to practice live Spanish conversation and get immediate corrections. Many students have part-time jobs, and the district recently bought new headsets. Which format fits this goal best?

- A. Short scheduled synchronous small-group sessions at several time options.
- B. Required three-hour live sessions at one fixed time.
- C. Asynchronous recorded lectures only.
- D. Buy more headsets.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A teacher at a school with a new video studio posts a 40-minute continuous lecture video with background music, animated borders, and unrelated stock clips. Students stop watching early. The teacher spent a weekend editing it. What change would most improve learning?

- A. Add more animations to hold attention.
- B. Break it into short segments students control and remove the music and unrelated clips.
- C. Keep it as is, since the teacher worked hard on it.
- D. Remove the video and post only the slides.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which video design best follows the segmenting principle?

- A. Five 4-minute clips, each on one step, with a Continue button.
- B. One 20-minute video with background music.
- C. One 20-minute video with no pauses.
- D. A 20-minute video with chapter titles that auto-play without stopping.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An online civics course, taught by a teacher who is also the debate coach, posts the prompt “What year was the Constitution signed?” All 30 posts give the same date and no one replies. The course runs on a new platform. Which prompt would spark discussion?

- A. Post the date and cite a source.
- B. Which constitutional compromise matters most today? Give one example and respond to a classmate who chose differently.
- C. Who signed the Constitution first?
- D. Hold a formal debate on the new platform.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Seventh graders researching climate find a polished website with a .org address, charts, and a professional logo. The school recently upgraded its Wi-Fi. What should the teacher coach students to do first to judge the site?

- A. Open new tabs and search what trusted sources say about the organization behind the site.
- B. Trust it, since .org sites are always reliable.
- C. Read the site's own About page and trust it.
- D. Judge it by how professional the charts look.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A student checks an unknown health website by searching the organization's name and finds several news reports that it is funded by a supplement company. What should she conclude?

- A. She should trust the site's About page instead.
- B. All health websites are unreliable.
- C. The site may be biased; she should seek independent sources.
- D. The site is reliable because it appears in the news.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher of an online history course, which launched last fall, notices many identical answers on multiple-choice quizzes. The course has 120 students and uses a free quiz tool. Which change best supports integrity?

- A. Assume all students cheat and give zeros.
- B. Rely on a lockdown browser alone.
- C. Switch to a paid quiz tool.
- D. Add tasks where students apply ideas to a local or personal case and explain their reasoning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher presents reading scores to families in a newly renovated media center. A bar chart shows Class A at 78 and Class B at 82, with the y-axis starting at 76, so Class B's bar looks three times as tall. What is wrong with the chart?

- A. Nothing; Class B scored three times higher.
- B. The chart should use a pie chart instead.
- C. The chart should start at 80 to be clearer.
- D. The y-axis should start at zero; the actual gap is only 4 points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

On a bar chart with the y-axis starting at 60, one bar shows 70 and another shows 90. How many times taller does the second bar appear?

- A. About 1.3 times
- B. 20 times
- C. 0.33 times
- D. 3 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher with five sections, who recently moved into a new classroom, tracks 140 students' quiz scores in a spreadsheet. She wants to quickly see which students are averaging below 70%. Which approach is most efficient?

- A. Scan the rows visually for low scores.
- B. Print all scores and sort them in the new classroom.
- C. Use an AVERAGE formula for each student and conditional formatting to highlight averages below 70.
- D. Calculate each average by hand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An online middle school science teacher, ~~whose course uses a new learning platform~~, posts weekly videos but never joins discussions or gives feedback beyond grades. Students say they feel alone. ~~The course has 90 students~~. Which presence is weakest?

Target word: classify the missing presence **Decides:** Teacher does not facilitate or connect.

A	Camouflage	The tool is not what is missing.
B	Correct	No facilitation or feedback; students feel isolated.
C	Adjacent category	The stem centers on facilitation and connection.
D	Absolute language	Teaching presence includes facilitation and feedback.

2. Answer C Concept 1 • transfer

Students in an online course post icebreaker introductions and reply warmly to each other. Which presence does this build most directly?

Target word: classify the presence **Decides:** Personal connection among learners.

A	Adjacent category	Not a presence in the framework.
B	Confusion set	Cognitive presence is inquiry and meaning-making.
C	Correct	Builds connection and trust.
D	Inverted operation	Teaching presence is design and facilitation.

3. Answer A Concept 2

A high school teacher of an online course wants students to practice live Spanish conversation and get immediate corrections. Many students have part-time jobs, and ~~the district recently bought new headsets~~. Which format fits this goal best?

Target word: sync vs. async choice **Decides:** Live conversation with immediate feedback.

A	Correct	Live interaction fits the goal; multiple times respect work schedules.
B	Absolute language	Ignores students' schedules.
C	Inverted operation	No live conversation or immediate correction.
D	Camouflage	Equipment does not decide format.

4. Answer B Concept 3

A teacher ~~at a school with a new video studio~~ posts a 40-minute continuous lecture video with background music, animated borders, and unrelated stock clips. Students stop watching early. ~~The teacher spent a weekend editing it~~. What change would most improve learning?

Target word: multimedia fix **Decides:** Long, unsegmented video with extraneous extras.

A	Inverted operation	Adds more extraneous material.
B	Correct	Segmenting and coherence reduce overload.
C	Camouflage	Effort does not decide effectiveness.
D	Dropped element	Drops the narrated explanation instead of fixing it.

5. Answer A Concept 3 • transfer

Which video design best follows the segmenting principle?

Target word: reverse: pick the segmented design **Decides:** Short, learner-paced chunks.

A	Correct	Short chunks at the learner's pace.
B	Confusion set	Adds extraneous sound; still not segmented.
C	Inverted operation	Continuous, not segmented.
D	Adjacent category	Labels without learner control.

6. Answer B Concept 4

An online civics course, ~~taught by a teacher who is also the debate coach~~, posts the prompt “What year was the Constitution signed?” All 30 posts give the same date and no one replies. ~~The course runs on a new platform.~~ Which prompt would spark discussion?

Target word: better prompt **Decides:** A single-answer factual prompt.

A	Adjacent category	Adds a source but still one answer.
B	Correct	Open-ended, requires reasoning and peer response.
C	Dropped element	Still a single factual answer.
D	Camouflage	Pulls in the coaching and platform details rather than fixing the prompt.

7. Answer A Concept 5

Seventh graders researching climate find a polished website with a .org address, charts, and a professional logo. ~~The school recently upgraded its Wi-Fi.~~ What should the teacher coach students to do first to judge the site?

Target word: digital literacy strategy **Decides:** An unfamiliar source.

A	Correct	Lateral reading checks the source from outside.
B	Absolute language	Domain alone does not prove credibility.
C	Adjacent category	Vertical reading relies on the source's self-description.
D	Confusion set	Design polish is not credibility.

8. Answer C Concept 5 • transfer

A student checks an unknown health website by searching the organization's name and finds several news reports that it is funded by a supplement company. What should she conclude?

Target word: reverse: interpret lateral reading results **Decides:** Outside sources show a conflict of interest.

A	Adjacent category	Returns to vertical reading.
B	Absolute language	Overgeneralizes.
C	Correct	Funding creates a conflict of interest.
D	Inverted operation	The news reports raise concerns, not credibility.

9. Answer D Concept 6

A teacher of an online history course, ~~which launched last fall~~, notices many identical answers on multiple-choice quizzes. The course has 120 students and ~~uses a free quiz tool~~. Which change best supports integrity?

Target word: integrity by design **Decides:** Easily copied recall questions.

A	Absolute language	Unfair and not evidence-based.
B	Dropped element	Surveillance alone does not address task design.
C	Camouflage	The tool is not the root problem.
D	Correct	Authentic, personalized tasks are hard to copy.

10. Answer D Concept 7

A teacher presents reading scores to families ~~in a newly renovated media center~~. A bar chart shows Class A at 78 and Class B at 82, with the y-axis starting at 76, so Class B's bar looks three times as tall. What is wrong with the chart?

Target word: misleading display **Decides:** Truncated axis exaggerates a 4-point gap.

A	Confusion set	Bar height misleads; 82 is not three times 78.
B	Adjacent category	Pie charts show parts of a whole, not comparisons of scores.
C	Inverted operation	Truncates even more.
D	Correct	$82 - 76 = 6$ vs. $78 - 76 = 2$ makes the bar 3× as tall; the real gap is 4 points.

11. Answer D Concept 7 · transfer

On a bar chart with the y-axis starting at 60, one bar shows 70 and another shows 90. How many times taller does the second bar appear?

Target word: compute visual distortion **Decides:** Heights above the baseline: 10 and 30.

A	Dropped element	That is the real ratio (90/70), ignoring the truncated axis.
B	Wrong stem number	Uses the raw difference.
C	Inverted operation	Divides the shorter bar by the taller.
D	Correct	$(90 - 60) / (70 - 60) = 30 / 10 = 3$.

12. Answer C Concept 8

A teacher with five sections, ~~who recently moved into a new classroom~~, tracks 140 students' quiz scores in a spreadsheet. She wants to quickly see which students are averaging below 70%. Which approach is most efficient?

Target word: productivity tool choice **Decides:** Many students; need automatic flagging.

A	Dropped element	Skips the calculation and is unreliable.
B	Camouflage	Irrelevant to efficiency.
C	Correct	Automates the calculation and flags students instantly.
D	Inverted operation	Slow and error-prone.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Community of Inquiry presences	Quality online learning combines teaching presence (design and facilitation), social presence (connection), and cognitive presence (building meaning through inquiry).	Thinking posted content alone creates teaching presence.	An online course feature to classify.
2	Synchronous vs. asynchronous	Use synchronous time for live interaction and quick feedback; use asynchronous work for flexible pacing and reflective thinking.	Assuming live sessions are always better or that everything should be recorded lectures.	A goal paired with a scheduling choice.
3	Segmenting and coherence in multimedia	Learners learn more from short, learner-paced segments and from media stripped of extraneous material.	Adding background music and decorations to make videos engaging.	A long, continuous video packed with extras.
4	Discussion prompts that invite interaction	Open-ended prompts that call for reasoning, examples, and responses to peers produce real discussion.	Asking a single factual question that everyone answers the same way.	Discussion posts that all repeat one answer.
5	Lateral reading for digital literacy	To judge an online source, leave it and check what other trusted sources say about it.	Judging credibility by how professional the site looks or its domain alone.	Students evaluating an unfamiliar website.
6	Integrity through assessment design	Authentic, applied, or personalized tasks reduce cheating in online assessment better than lockdown tools alone.	Relying only on surveillance to protect integrity.	Concerns about copying on online quizzes.
7	Honest data displays	Bar charts should start at zero and use consistent scales so differences are not exaggerated.	Trusting a chart whose truncated axis makes small gaps look huge.	A chart where the y-axis starts well above zero.
8	Spreadsheets for productivity	Formulas and conditional formatting automate calculations and flag students who need support.	Calculating by hand or scanning rows visually.	A teacher tracking scores for many students.

Course scope (public description): Technology for Instruction and Online Pedagogy is an innovative course designed to equip educators with the skills to effectively integrate technology in their teaching practices. The course also covers best practices for online pedagogy, assessment and feedback, collaborative learning, and the use of multimedia and interactive elements to enhance learning experiences. With a focus on practical application, educators will leave the course ready to create and facilitate compelling, high-quality online learning experiences that meet the needs of today's learners. This course also provides a foundation for supporting digital literacy in K-12 education. In addition, this course prepares candidates to use technology to improve professional productivity and effectiveness in areas like data analysis and data representations. Candidates will apply evidence-based practices to adapt instruction to meet student needs.